1 The nature of technology: Our computers

You will learn

- what are the main parts of a computer
- what the main parts of a computer are for
- things computers can and cannot do.

Computers are all around us.

Computers are in our phones. Computers are in our homes.

Computers can help us learn and do our work.

You need to learn to use computers safely and in ways that make the world a better place.





Talk about...

Why do you think it is important to know how a computer works?



Learning outcomes: Name the main parts of a typical computer and what they are for; Say some things a computer can do and some things a computer cannot do



Class activity

Look at a computer in your classroom. Draw the computer on a big piece of paper. How many parts have you drawn? Can you name the parts? Talk to a partner about the names you know. Write the names on the drawing.

input
output device
mouse processor
wireless screen



Did you know?

Computers use electricity to do their work. There are many ways to make electricity. Renewable electricity means electricity from sources that will never run out such as sunshine or wind power.



1.1 Computers are electric

In this lesson

You will learn:

- what a processor is
- how computers use electricity.

Spiral back

In Student Book 1 you learned what a computer is. In this unit you will learn what is inside a computer.

What is electricity?

A **device** is something people make to do a useful task. Many devices use electricity.

Electricity is a kind of energy. Wires bring electricity. The electricity flows along the wire.

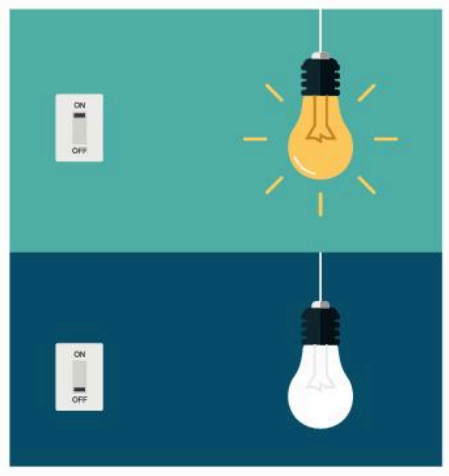
But electricity is dangerous. If you touch an electric wire you will get a bad shock. It will hurt you. So be very careful with anything that has electricity in it.



Electricity ON and OFF

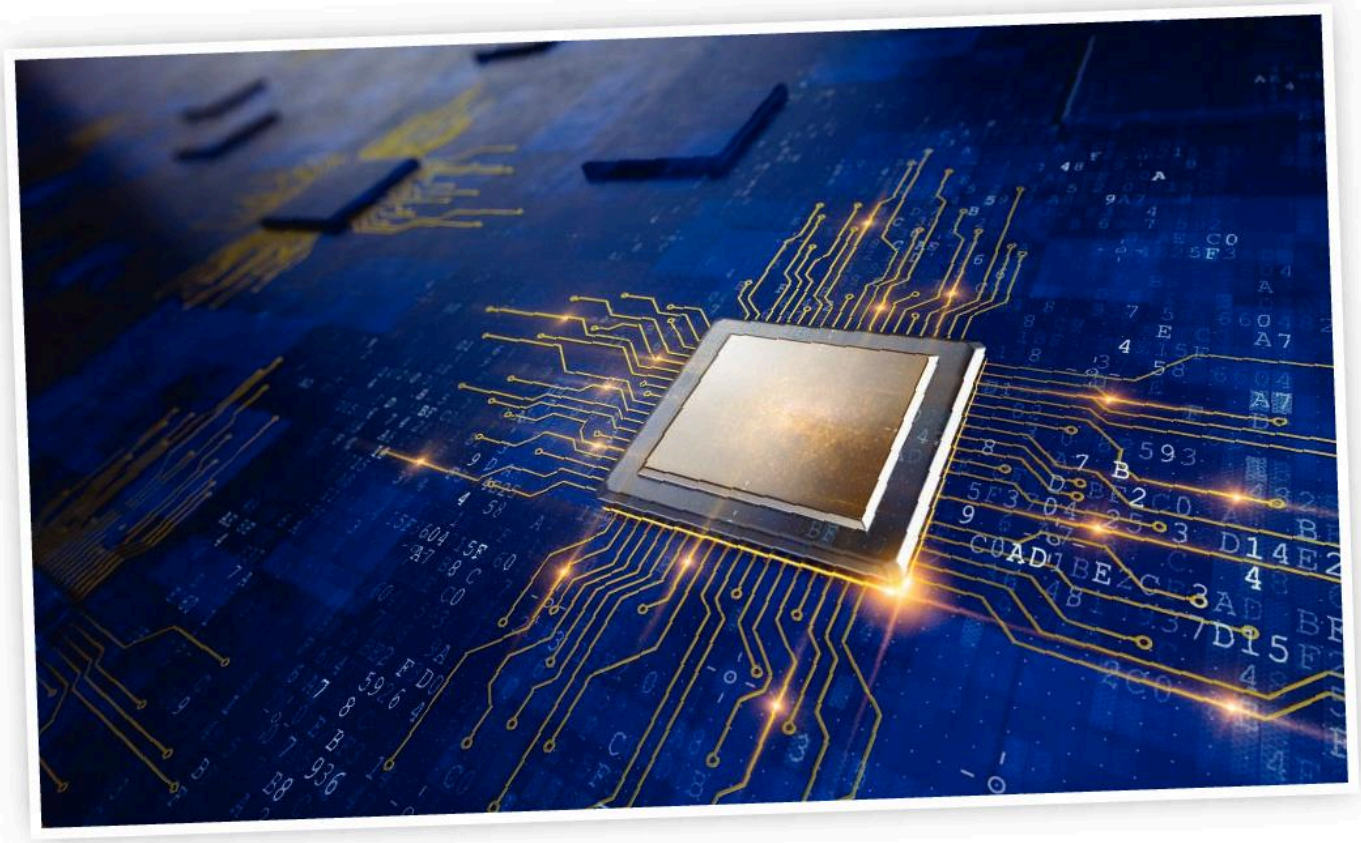
Electricity can be ON or OFF. Think of a light bulb. When the switch is ON, electricity can get through. The light goes on.

The OFF switch stops the electricity. The light goes off.



Inside the computer

There is an electrical device inside the computer. It is called a **processor**. The processor uses on/off electrical signals. Everything that happens inside the processor is made of on/off electrical signals.



Activity

Tell a partner about the last time you used a computer. What did you use it for?

Extra challenge

Draw a computer. Show the wire that brings electricity to the computer. Draw a circle where you think the processor is.

Think again

What must you do to stay safe when you use electronic devices?

1.2 Output

In this lesson

You will learn:

- what output is
- what computer parts are used for output.

Inside a computer

A computer holds lots of information. For example numbers and pictures. It holds the information using electricity.

But you cannot see the electricity. The circuits are too small. It is dangerous to open a computer when the electricity is switched on.

Output devices

You cannot look at the electricity inside a computer. How can you find out what is inside the computer?

You use an **output device**.

Output devices take the electricity from the processor. They turn the electrical signals into things you can see and use.

Output you can see

The **screen** or **monitor** of a computer is an output device. It takes information from the processor. It turns it into colours and shapes.



A monitor makes visual output. That means you can see the output.



A printer gives you output. The output is on paper. You can keep the output after the computer is switched off.

Other types of output

There are other types of output.





- **Sound output:** Speakers and headphones make sound output.
- **Movement:** A computer can make a machine move. For example, a machine can clean the floor.



Activity

What devices are used for output? Draw two different output devices. Write the names.

Extra challenge

A 3D printer makes solid objects. For example, a 3D printer can stick blobs of plastic together. See if you can find out more.

Think again

A monitor makes visual output. Say one other type of output.